MODELING THE THOMSON PROBLEM WITH CIRCLE PACKINGS

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Abstract. The Thomson problem is longstanding conjecture with applications in physics and biology. Number 7 on Smale's list of conjectures, the Thomson problem asks how to find the minimum energy distribution of N electrons on the surface of the sphere. In general, finding this minimum is an NP hard problem. Thus progress on this problem must rely on computational methods. In this project, we use circle packings to construct a discrete model of the Thomson problem. Circle packings of minimum configurations are experimentally found to be good approximations and appear to converge to the actual energy. Furthermore, we demonstrate how these models can be combinatorial manipulated, and conjecture how this might be used to compute energy minimums.

The Thomson Problem

The Thomson problem asks how to find the minimum energy distribution of N electrons on the surface of the sphere. The *energy* of N points, $p_1, p_2, \ldots, p_N$, on the sphere is defined as **?**

$$E = \sum_{i<j} \frac{1}{|p_i - p_j|}.$$

Number 7 on Smale's list of conjectures, this problem remains one of the most important unsolved packing problems. Not just important in mathematics, the Thomson problem is related or equivalent to many other problems in biology, physics, computer science, chemistry, etc **?**.

Known solutions have been rigorously proven for only a small number of special cases. In general, finding a solution is an NP hard problem [need reference], and progress has relied on computational methods. Using such methods, smallest known energy distributions for most points sets less than 1000 have been found**?**.

Circle Packing

Circle packings are configurations of circles with specified patterns of tangency. The following theorem provides rigidity and existence of circle packings on the sphere.

Koebe-Andreev-Thurston Theorem. *There exists a circle packing on the sphere for any finite triangulation of the sphere. Furthermore, this circle packing is unique up to Möbius transformations.*

The actual radii for the circle packing guaranteed by the theorem above can always be found using the *Circle Packing Algorithm***?**. Moreover we have a feature rich software package, `CirclePack`[1], for computing and visualizing the lovely results.

Circle packings naturally express a conformal geometry which has allowed the field of circle packing to be developed into a rich theory of discrete analytic functions. Many classical complex functions can be approximated using circles packings, most famously the Riemann mapping theorem:

Thurston's Circle Packing Conjecture Theorem. *The conformal mapping of a bounded simply connected domain onto the unit disc can be approximated with a circle packing.*

In summary, we have the following

Thanks to HBCU-UP of Johnson C. Smith for funding this project.

Thanks to Ken Stephenson for serving as an advisor on this project.

[1] www.math.utk.edu/ kens/CirclePack; special thanks to Ken Stephenson for developing and maintaining this freely available software

Figure 0.1. A tangency pattern and its corresponding circle packing.

Thomson_problem_HBCUP_grant_project/images/graph_minimal.pdf

FIGURE 0.2. Error

(1) An essentially unique circle packing for a spherical triangulation exists.
(2) The circle packing can be computationally realized using `CirclePack`.
(3) As the number of circles increase, a circle packing analogue will converge towards its continuous counterpart**?**.

Our goal

Repelling points on a flat surface naturally adopt a hexagonal pattern. However the topology of the sphere precludes a triangulation where every vertex has 6 neighbors. Try wrapping a sheet of paper around a basketball –imperfections in your wrapping job are unavoidable.

The Euler characteristic states that there will be 12 such "imperfections" where a vertex has 5 neighbors. The truncated icosahedron, a soccer ball, is a classic example. Thus a combinatorial perspective may shed new light on this problem.

We first experimentally demonstrated that circle packings are a good model of the Thomson problem. Then we demonstrate how manipulating the combinatorics of circle packings might be used to energy.

A discrete Model of the Thomson Problem

The Thomson problem is well studied, and there is a large catalog of current minimal energy distributions**?**. These point sets are stored in the interactive web applet, `thomsonapplet`[2], which can also generate random point sets, measure the energy, and implement a wide array of algorithms.

To construct a circle packing model for N energy points on the sphere we did the following:

(1) Find the Delaunay triangulations for N points using data from `thomsonapplet`.
(2) Compute the circle packing for this triangulation using `CirclePack`.
(3) Normalize the packing by applying a Möbius map such that the centroid of the circle centers in $\mathbb{R}^3$ is at the origin.

Measure the *error* of this model as,

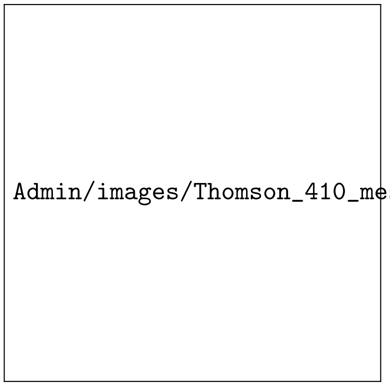
$$error = \frac{\text{energy from circle packing-energy}}{\text{energy}}$$

Figure (XX) shows the error for minimal known distribution for N point. The error from our experiments never exceeded 0.0062. As expected the error tended toward 0 for large values of N. It appears that the error introduced by the discreteness of circle packings is limited for even small N and inconsequential for larger N.

Combinatorial Manipulaitons

A Whitehead move, or "flip", replaces one interior edge with another in the union of two edge-sharing faces (see figure XXX). Such a modification does not change the number of points -only the combinatorics locally. However these local changes result in a new circle packing and thus invoke a global geometric change. A common issue with optimization algorithms is settling into a local but not a global minimum. In the figure below (figure YYY), a two defects are separated be an area of degree six vertices. Following a sequence of flips, the defects are then combined and neutralized.

[2]http://thomson.phy.syr.edu, Cris Cecka, Primary Author.



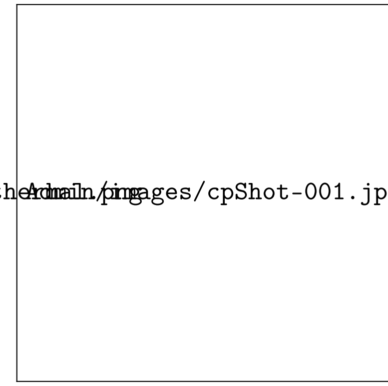


Figure 0.3. An energy distribution modeled with circle packing.